

COMP9334 Solution to the revision problems for Week 8B

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Question 1

- (a). The decision variables are W_1 and W_2 . The linear programming problem is:

$$\min R$$

subject to

$$\begin{aligned} 3 + 5W_1 &\leq R \\ 1 + 7W_2 &\leq R \\ W_1 &\geq W_{1,\text{base}} \\ W_2 &\geq W_{2,\text{base}} \\ W &= W_1 + W_2 \end{aligned}$$

- (b). The files `hw_lp.mod` and `hw_batch` are available at the course website. The workload allocation should be $W_1 = 3.3333$ and $W_2 = 2.6667$.
- (c). If $W_1 = \frac{10}{3}$, then the response time of Server 1 is $3 + 5 \times \frac{10}{3} = \frac{59}{3}$. If $W_2 = \frac{8}{3}$, the response time of Server 2 is $1 + 7 \times \frac{8}{3} = \frac{59}{3}$. You will notice that the response times of the two servers are identical.

In order to explain why the response times are identical, let us work out some examples to understand the problem a bit more.

Let us say if we allocate 3 units of workload to each server. The response time of Server 1 is $3 + 5 \times 3 = 18$. The response time of Server 2 is $1 + 7 \times 3 = 22$. The system response time is the bigger of 18 and 22, which is 22.

Since our goal is to minimise the system response time, let us see how we can reduce the system response time. Since both response times are strictly increasing function of the workload, you can increase (resp. decrease) the response time by increasing (decreasing) the workload. Since Server 2 has the higher response time, it determines the system response time. (Think about bottleneck!) Therefore, if we want to try to reduce the system response time, we should try to reduce the response time of Server 2. This can be done by shifting some workload from Server 2 to Server 1. Let us try to decrease the response time of Server 2 by shifting some workload to Server 1. Let us try $W_1 = 3.1$ and $W_2 = 2.9$. With this allocation, we have:

response time of Server 1 is $3 + 5 \times 3.1 = 18.5$ and the response time of Server 2 is $1 + 7 \times 2.9 = 21.3$. The system response time is the bigger of 18.5 and 21.3, which is 21.3.

The example in the last paragraph shows that we can decrease the system response time by shifting some workload from Server 2 to Server 1. We can keep doing that until the response time of the two servers are the same. When this happens, we can no longer decrease the system response time any longer because any shift in the workload between the servers will increase the system response time. This is the minimum system response time.

- (d). To achieve the minimum system response time, W_1 should be 3.2 and W_2 should be 2.8.
- (e). If $W_1 = 3.2$, then response time of Server 1 is $3 + 5 \times 3.2 = 19$. If $W_2 = 2.8$, the response time of Server 2 is $1 + 7 \times 2.8 = 20.6$. Server 2 has the higher response time so it determines the system response time. If you want to further reduce the response time, you need to shift some workload from Server 2 to Server 1. However, the workload on Server 2 is the minimum allowed, so we cannot shift any more workload from Server 2 to Server 1.